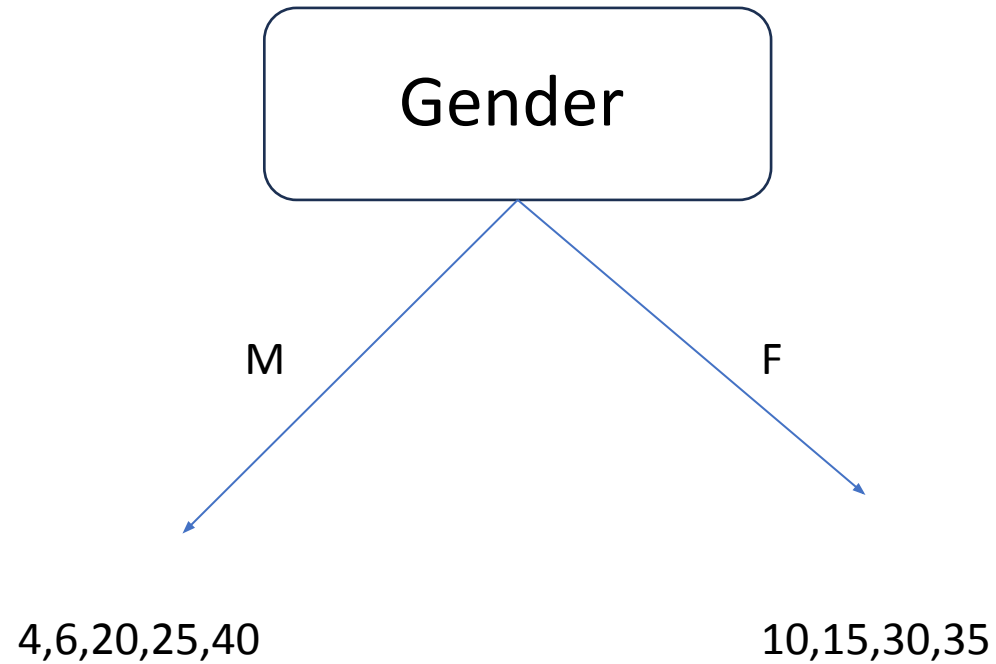


Step by Step Procedure for Regression Tree Construction

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0



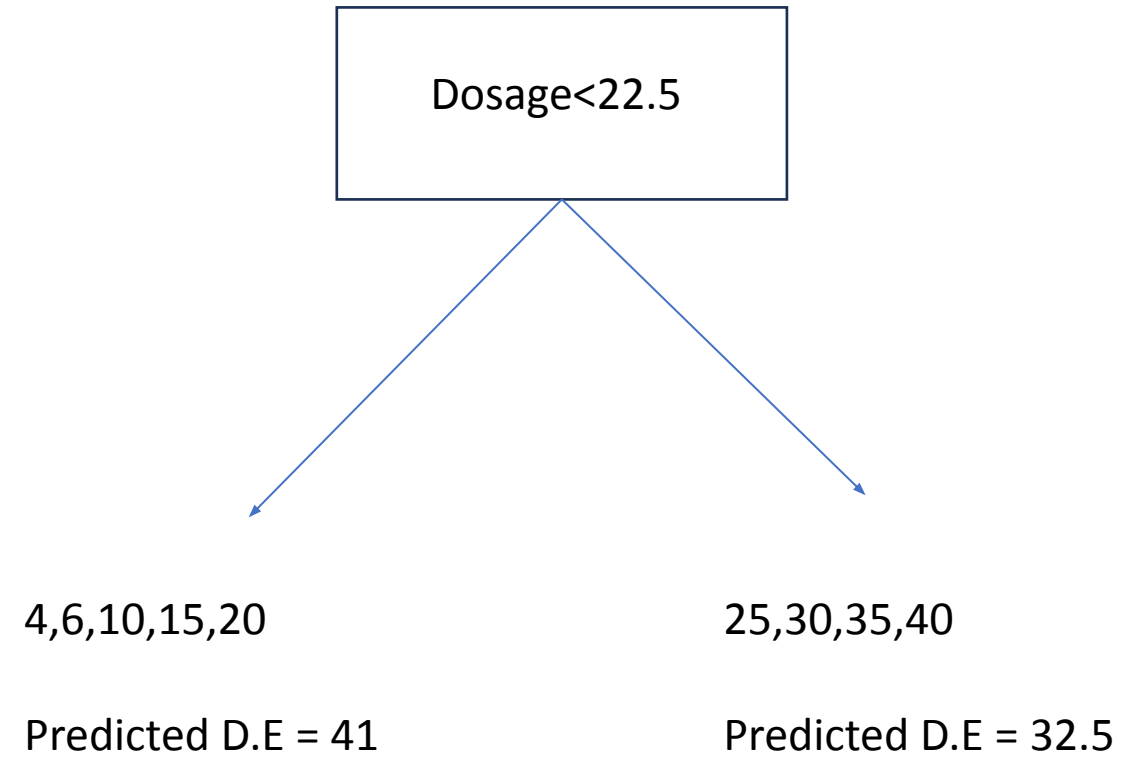
$$\text{Predicted D.E} = (0+2+100+100+0)/5 \\ = 40.4$$

$$\text{Predicted D.E} = (3+100+20+10)/4 \\ = 33.25$$

$$\text{SSR (Gender)} = (0-40.4)^2 + (2-40.4)^2 + (100-40.4)^2 + (100-40.4)^2 + (0-40.4)^2 \\ + (3-33.25)^2 + (100-33.25)^2 + (20-33.25)^2 + (10-33.25)^2 \\ = 17929.95$$

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Threshold
5
8
12.5
17.5
22.5
27.5
32.5
37.5



$$\begin{aligned}
 \text{SSR (Dosage} < 22.5) &= (0-41)^2 + (2-41)^2 + (3-41)^2 + (100-41)^2 + (100-41)^2 \\
 &\quad + (100-32.5)^2 + (20-32.5)^2 + (10-32.5)^2 + (0-32.5)^2 \\
 &= 17883
 \end{aligned}$$

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

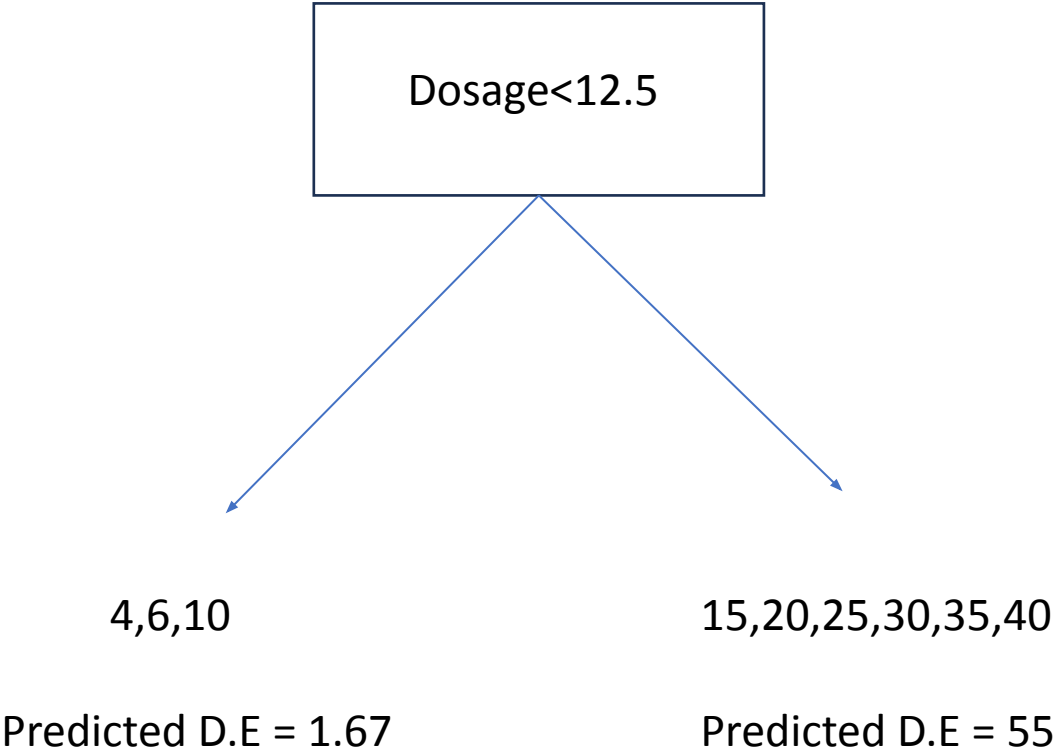
Threshold
5
8
12.5
17.5
22.5
27.5
32.5
37.5

SSR
16485
14670
12355
17177
17883
14709
15374
16485

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Threshold
5
8
12.5
17.5
22.5
27.5
32.5
37.5

SSR
16485
14670
12355
17177
17883
14709
15374
16485



Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Threshold

17.5

22.5

27.5

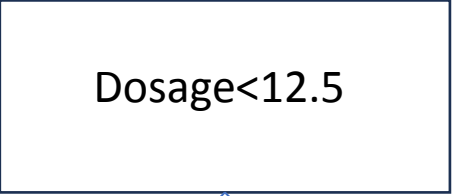
32.5

37.5

SSR

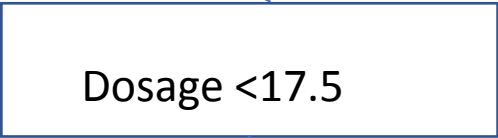
9920

6275



4,6,10

Predicted D.E = 1.67



15

Predicted D.E = 100

20,25,30,35,40

Predicted D.E = 46

$$\text{SSR}(\text{Dosage} < 17.5) = (100-100)^2 + (100-46)^2 + (100-46)^2 + (20-46)^2 + (10-46)^2 + (0-45)^2 = 9920$$

$$\text{SSR}(\text{Dosage} < 22.5) = (100-100)^2 + (100-100)^2 + (100-32.5)^2 + (20-32.5)^2 + (10-32.5)^2 + (0-32.5)^2 = 6275$$

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Threshold

SSR

17.5

9920

22.5

6275

27.5

200

32.5

7425

37.5

13076

SSR(Dosage<27.5) =

$(100-100)^2 + (100-100)^2 + (100-100)^2 + (20-10)^2 + (10-10)^2 + (0-10)^2 = 200$

$SSR(32.5) = (100-105)^2 + (100-105)^2 + (100-105)^2 + (20-105)^2 + (10-5)^2 + (0-10)^2 = 7425$

$SSR(37.5) = (100-66)^2 + (100-66)^2 + (100-66)^2 + (20-66)^2 +$

Dosage<12.5

4,6,10

Predicted D.E = 1.67

15,20,25,30,35,40

Predicted D.E = 55

Dosage <27.5

15,20,25

Predicted D.E = 100

30,35,40

Predicted D.E = 10